- **1. Select a mathematical description or a representation that describes a problem**
  - **Analysis:** This is the core requirement for solving the problem. Students need to choose a formula from four options that **correctly** describes the mathematical model of the penguin population changing over time, based on the problem description (birth rate, death rate, time). This involves the ability to translate a real-world scenario into an abstract mathematical expression.
- **5. Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations**
  - **Analysis:** Students need to recognize that this is an **exponential growth/decay** model (rather than a linear model), where the population changes by a constant compound growth factor each year. Recognizing this "change and relationship" mathematical structure is crucial for selecting the correct exponential form (with an exponent of 7).
- **4. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation**
  - **Analysis:** This ability emphasizes understanding the textual assumptions and converting them into mathematical components. For example, understanding that "1 chick per couple" means the adult population increases by 50%, or a multiplier factor of 1 + 0.5 = 1.5. Understanding that "20% will die" means an 80% survival rate, or a multiplier factor of 0.8. Accurately decoding this information is a prerequisite for building the correct model.
- **6. Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables**
  - **Analysis:** Students need to identify the key variables: initial population (10,000), time frame (7 years), and the combined annual change factor (1.5 × 0.8).
- **9. Translate a problem into a standard mathematical representation or algorithm**
  - **Analysis:** Students must combine the identified growth and decay factors into the form of a standard exponential growth formula:
    $P = \text{Initial Value} \times (\text{Annual Growth Factor})^{\text{Years}}$, which is
    $P = 10\,000 \times (1.5 \times 0.8)^7$.

- **1. Select a mathematical description or a representation that describes a problem** (p. 1)
  - **分析：** 这是解决问题的核心要求。学生需要根据题目描述（出生率、死亡率、时间）从四个备选公式中选择一个**正确**地描述企鹅数量随时间变化的数学模型。这涉及到将现实情境转化为抽象的数学表达能力。
- **5. Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations**
  - **分析：** 学生需要识别出这是一个**指数增长/衰减**的模型（而非线性模型），其中人口每年以恒定的复合增长因子变化。识别这种“变化和关系”的数学结构对于选择正确的指数形式（带有指数7）至关重要。
- **4. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation** (p. 1)
  - **分析：** 这项能力强调理解文本假设并将其转化为数学组件。例如，理解“每对夫妇1只小企鹅”意味着成年企鹅数量增加50%，即乘数因子为 1 + 0.5 = 1.5。理解“20%死亡”意味着存活率为 80%，即乘数因子为 0.8。准确解码这些信息是构建正确模型的前提。
- **6. Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables**
  - **分析：** 学生需要识别出关键变量：初始数量（10,000）、时间（7年）以及每年变化的综合因子（1.5 × 0.8）。
- **9. Translate a problem into a standard mathematical representation or algorithm**
  - **分析：** 学生必须将被识别出的增长和衰减因子组合成一个标准指数增长公式的形式：$P = \text{初始值} \times (\text{年增长因子})^{\text{年数}}$，即 $P = 10\,000 \times (1.5 \times 0.8)^7$。